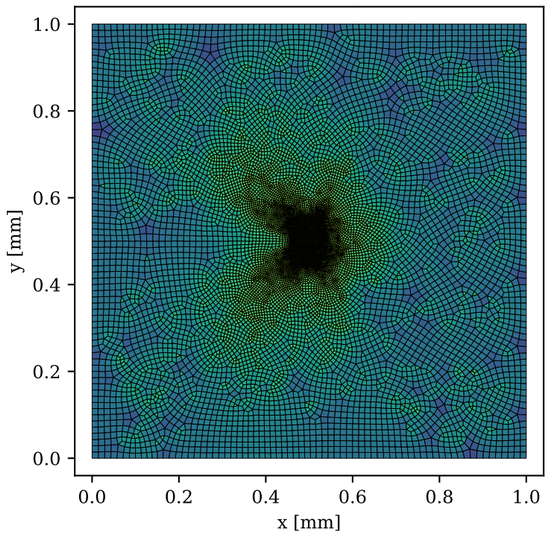
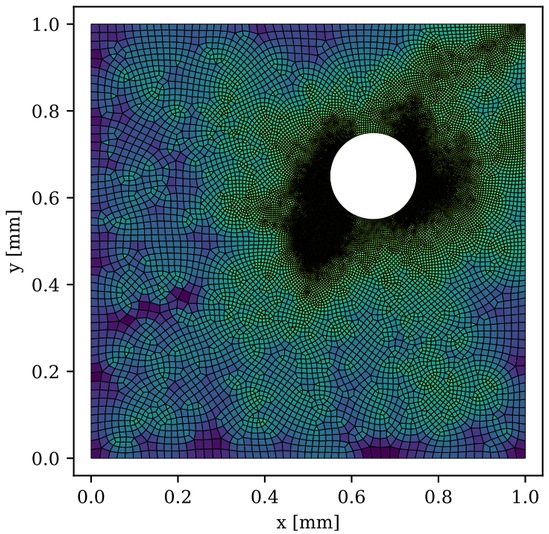
F1736: Independent Scientific Reassessment of Five-Geometry Method A Adaptive Meshes
(Authentic Final Meshes from Tasks F1725, F1726, F1728, F1734, and Task 192)

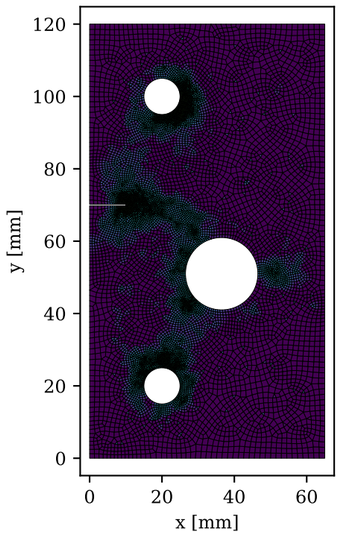
(a) Case 1: Mode-I Single-Edge Notched Plate
Authoritative: 14147 el (97.3% Q) | Pub: 13762 el
$\ell_0 = 0.0075$ **mm** | $h_{min} = 0.59\,\mu$**m** | $h \le \ell_0$**: 55.8%**



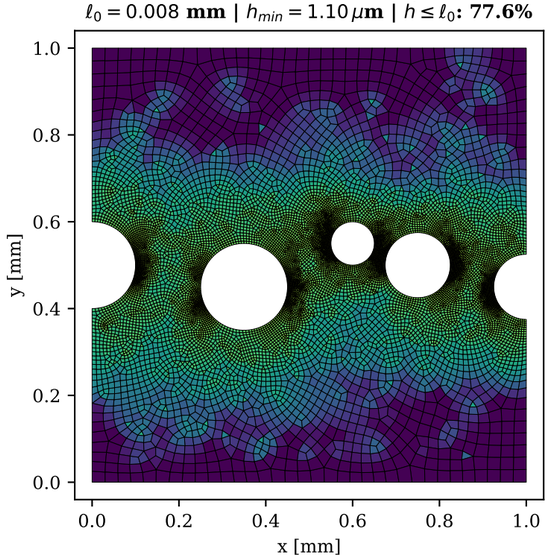
(b) Case 2: Plate with an Off-Axis Hole
Authoritative: 23965 el (97.2% Q) | Pub: 23618 el
$\ell_0 = 0.0075$ **mm** | $h_{min} = 0.66\,\mu$**m** | $h \le \ell_0$**: 79.6%**



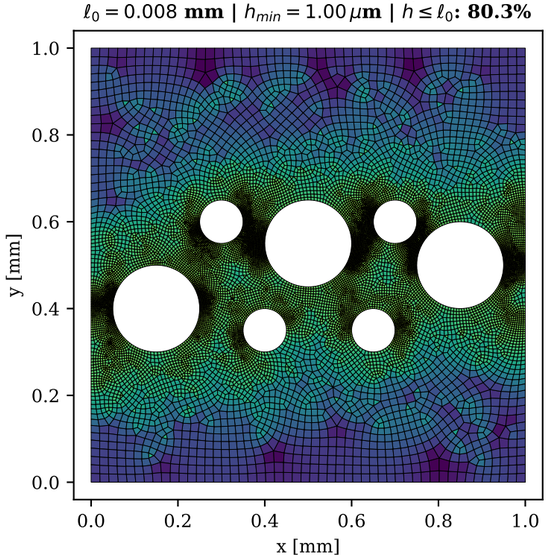
(c) Case 3: Rectangular Notched Plate with Hole and Pins
Authoritative: 12110 el (96.7% Q) | Pub: 12168 el
$\ell_0 = 0.285$ **mm** | $h_{min} = 80.36\,\mu$**m** | $h \le \ell_0$**: 16.5%**



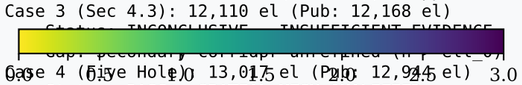
(d) Case 4: Five-Hole Plate
Authoritative: 13017 el (97.5% Q) | Pub: 12944 el
$\ell_0 = 0.008$ **mm** | $h_{min} = 1.10\,\mu$**m** | $h \le \ell_0$**: 77.6%**



(e) Case 5: Seven-Hole Plate
Authoritative: 15708 el (97.6% Q) | Pub: 15714 el
$\ell_0 = 0.008$ **mm** | $h_{min} = 1.00\,\mu$**m** | $h \le \ell_0$**: 80.3%**




```
TASK F1736: AUDITED BENCHMARK STATUS
-----------------------------------------------------
Case 1 (Mode I): 14,147 el (Pub: 13,762 el)
  - Status: PARTIAL — LOCALIZED REFINEMENT
  - Gap: Ligament distal to tip unrefined prior to crack
Case 2 (Off-Axis Hole): 23,965 el (Pub: 23,618 el)
  - Status: PASS — SPATIAL MESH QUALIFIED
  - Note: Fracture sim ran (252 inc), stopped at min dt
Case 3 (Sec 4.3): 12,110 el (Pub: 12,168 el)
  - Status: INCONCLUSIVE — INSUFFICIENT EVIDENCE
  - Gap: Secondary corridor unrefined (h > ell_0/
Case 4 (Five Hole): 13,017 el (Pub: 12,944 el)
  - Status: PASS — SPATIAL MESH QUALIFIED
  - Note: Fracture sim NOT DEMONSTRATED
Case 5 (Seven Hole): 15,708 el (Pub: 15,714 el)
  - Status: PASS — SPATIAL MESH QUALIFIED
  - Note: Fracture sim NOT DEMONSTRATED
-----------------------------------------------------
*Zero negative Jacobians across all 5 meshes.
```



Spatial Resolution Ratio (ℓ_0/h) (Capped at 3.0)